

* Rank 0 tensor.

$$[Q]$$

$$Q = \mathbb{Q}_0^{(1)}$$

* Rank 1 tensor.

$$[0 \quad 0 \quad Q_z]$$

$$Q_z = \mathbb{Q}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} Q_{xx} & 0 & 0 \\ 0 & Q_{xx} & 0 \\ 0 & 0 & Q_{zz} \end{bmatrix}$$

$$Q_{xx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(1)}$$

$$Q_{zz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & Q_{xxz} \\ 0 & 0 & Q_{xxz} \\ 0 & 0 & Q_{zzz} \\ 0 & Q_{xxz} & 0 \\ Q_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxz} = \mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{\alpha(1)}$$

$$Q_{zzz} = 3\mathbb{Q}_z^{(1)} + 2\mathbb{Q}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & Q_{xxz} \\ 0 & 0 & Q_{xxz} \\ 0 & 0 & 0 \\ 0 & -\frac{Q_{xxz}}{2} & 0 \\ -\frac{Q_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxz} = 2\mathbb{Q}_z^{(2)}$$

$$\frac{Q_{xxz}}{2} = \mathbb{Q}_z^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & Q_{yzy} & 0 \\ -Q_{yzy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{yzy} = \mathbb{Q}_z^{(3)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & 0 & 0 \\ Q_{xxyy} & Q_{xxxx} & Q_{xxzz} & 0 & 0 & 0 \\ Q_{xxzz} & Q_{xxzz} & Q_{zzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{yyzz} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{yyzz} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$\begin{aligned} Q_{xxxx} &= \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} \\ Q_{xxyy} &= \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} \\ Q_{xxzz} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} \\ Q_{zzzz} &= \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)} \\ Q_{yyzz} &= \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} \\ Q_{xyxy} &= \mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(2)} \end{aligned}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & Q_{xxzz} & 0 & 0 & 0 \\ 0 & 0 & Q_{xxzz} & 0 & 0 & 0 \\ -Q_{xxzz} & -Q_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxzz} = 3\mathbb{Q}_u^{(3)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} Q_{yyzz} & 0 & 0 \\ 0 & Q_{yyzz} & 0 \\ 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$\begin{aligned} Q_{yyzz} &= \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} \\ Q_{xyxy} &= \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)} \end{aligned}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ Q_{yyzz} & 0 & 0 \\ 0 & -Q_{yyzz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{yyzz} = -3\mathbb{Q}_u^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yyzz} & 0 & 0 \\ 0 & 0 & 0 & 0 & -Q_{yyzz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$Q_{yyzz} = -3\mathbb{Q}_u^{(6)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ Q_{xxyy} & Q_{xxxx} & Q_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ Q_{zzxx} & Q_{zzxx} & Q_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & Q_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{yzzy} & 0 & 0 & Q_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{xyxy} & 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$\begin{aligned} Q_{xxxx} &= \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} \\ Q_{xxyy} &= \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} \\ Q_{xxzz} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} \\ Q_{zzxx} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} \\ Q_{zzzz} &= \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)} \\ Q_{yzyz} &= \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} \\ Q_{yzzy} &= \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} \\ Q_{xyxy} &= \mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(2)} \end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & Q_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & -Q_{yzzy} & 0 & 0 & -Q_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{xyxy} & 0 & 0 & -Q_{xyxy} \end{bmatrix}$$

$$\begin{aligned} Q_{yzyz} &= \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)} \\ Q_{yzzy} &= -\mathbb{Q}_0^{(3)} + 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)} \\ Q_{xyxy} &= \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} & Q_{xxxx} & & 0 & 0 \\ & 0 & & Q_{xxxx} & 0 \\ & 0 & & 0 & Q_{zzzz} \\ & 0 & & 0 & Q_{yyzz} \\ & Q_{zzxx} & & 0 & 0 \\ & 0 & & Q_{xxyy} & 0 \\ & 0 & & Q_{yzzy} & 0 \\ & 0 & & 0 & Q_{zzxz} \\ Q_{xxyy} + Q_{yyzz} - Q_{yzzy} - Q_{zzxz} + Q_{zzxx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} Q_{xxxx} &= 3\mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} \\ Q_{zzzz} &= 3\mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)} \\ Q_{yyzz} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} \\ Q_{zzxx} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} \\ Q_{xxyy} &= \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} \\ Q_{yzzy} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} \\ Q_{zzxz} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} \end{aligned}$$